

# Ruichen Xu

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 [LinkedIn](#) |  [ru1ch3n](#) |  [Google Scholar](#) | [OpenReview](#)

## SUMMARY

Ph.D. candidate in Applied Mathematics & Statistics at Stony Brook University. Research interests: **predictive representation learning (JEPA)**, **neural operators** for sparse PDE inference, **physics-aware generative models**, and **LLM-guided molecular optimization**. I build reproducible pipelines and run large-scale experiments on SLURM/CUDA HPC clusters.

## EDUCATION

- **Stony Brook University (The State University of New York)** Aug 2022 – Present (Expected 2027)  
*Ph.D., Computational Applied Mathematics (GPA 4.00/4.00)* Stony Brook, NY, USA
- **Courant Institute of Mathematical Sciences, New York University (NYU)** Sep 2020 – May 2022  
*M.S., Mathematics (GPA 4.00/4.00)* New York, NY, USA
- **University of California, Davis** Sep 2019 – Jun 2020  
*M.S., Statistics (GPA 3.94/4.00)* Davis, CA, USA
- **Beijing University of Chemical Technology (BUCT)** Sep 2015 – Jun 2019  
*Bachelor, Financial Mathematics (89.33/100); Minor: Commercial Management* Beijing, China

## PROFESSIONAL EXPERIENCE

- **AI3, Stony Brook University** Jun 2026 – Jul 2026  
*REU Research Mentor* Stony Brook, NY, USA
  - Mentored REU students on AI4Science research projects at AI3, including project scoping, literature review, experiment design, and technical implementation.
  - Guided students on reproducible research workflows, code organization, experiment tracking, and presentation preparation.

## PROJECTS

- **JEPA for PDEs and Graph-Structured Scientific Data** Jan 2026 – Present  
*Tools: Python, PyTorch; Topics: JEPA, PDE dynamics, graph learning, self-supervised learning, AI4Science*
  - **HP-JEPA**: Uses hierarchical JEPA to learn multi-scale graph representations. [\[arXiv\]](#)
  - **JENO**: Uses full-field latent prediction for inverse PDE recovery from sparse observations.
  - **SCOPE**: Combines latent prediction and physical grounding for sparse PDE recovery. [\[GitHub\]](#)
- **PDE-OBS / PartialObs–PDEBench: Partial-Observation Benchmark & AI4Physics Collection** 2025 – Present  
*Tools: Python; Topics: partial observations, benchmarking, diffusion, solver recommendation* [\[Project page\]](#)
  - Maintains **AI4Physics**: curated AI4PDE/AI4SDE collection (taxonomy + tags + takeaways) and hosts ongoing results.
  - Develops **PDE-OBS** for controlled evaluation of PDE learning under sparse, irregular, sensor-style, and masked observation regimes.
  - Builds diffusion-based conditional samplers (APOD/APOD++) for reconstructing and forecasting PDE states from partial observations with unified metrics.
  - Organizes benchmark metadata, reproducible configs, and evaluation scripts to compare neural operators, diffusion models, and physics-aware baselines.
- **ORACLE: LLM-Guided Edits + Simulated Annealing for Structure-Based Drug Design (SBDD)** 2025 – Present  
*Tools: Python, PyTorch; Topics: LLM-guided molecular editing, multi-objective optimization*
  - Combines LLM-proposed chemistry-aware edits with simulated annealing to explore local molecular neighborhoods while enforcing validity constraints.
  - Optimizes multi-objective scores including docking, QED, SA-score, diversity, and constraint satisfaction, with configurable objective weights.
  - Implements ablation and tracking pipelines for edit operators, temperature schedules, acceptance rules, and Pareto trade-off analysis.
  - Evaluates controllable molecular optimization through optimization curves, chemically valid generation, and Pareto trade-offs.

## PUBLICATIONS

### Peer-Reviewed – Conferences

- [c.1] **GAGA: Gaussianity-Aware Gaussian Approximation for Efficient 3D Molecular Generation.** Jingxiang Qu, Wenhan Gao, Ruichen Xu, Yi Liu. *ICLR* 2026. ([OpenReview](#))
- [c.2] **Velocity-Inferred Hamiltonian Networks: Symplectic Dynamics from Position-Only Observations.** Ruichen Xu, Claire Yu, Zongyu Wu, Siyao Wang, Luoyao Chen, Georgios Kementzidis, Haochun Wang, Yuefan Deng. *NYSDS 2025, SIAM Proceedings*. ([SIAM DOI](#))
- [c.3] **Discretization-invariance? On the Discretization Mismatch Errors in Neural Operators.** Wenhan Gao, Ruichen Xu, Yuefan Deng, Yi Liu. *ICLR* 2025. ([OpenReview](#)) ([ICLR Page](#))
- [c.4] **Kolmogorov–Arnold Representation for Symplectic Learning: Advancing Hamiltonian Neural Networks.** Z. Wu, R. Xu<sup>†</sup>, L. Chen, G. Kementzidis, S. Wang, Y. Deng. *IJCNN* 2025. ([IEEE](#)); <sup>†</sup>co-first author; project lead.
- [c.5] **Boundary-Informed Method of Lines for Physics-Informed Neural Networks.** Maximilian Cederholm, Siyao Wang, Haochun Wang, Ruichen Xu, Yuefan Deng. *NYSDS 2025, SIAM Proceedings*. ([SIAM DOI](#)); co-corresponding author (with Y. Deng).
- [c.6] **Physics-Informed Active Learning via Functional Simulated Annealing for Neural Operator.** Albert Ding, Siyao Wang, Haochun Wang, Ruichen Xu, Yuefan Deng. *NYSDS 2025, SIAM Proceedings*. ([SIAM DOI](#)); co-corresponding author (with Y. Deng).

### Peer-Reviewed – Journals

- [j.1] **Towards Principled Test-Time Adaptation for Time Series Forecasting.** Haochun Wang, Ruichen Xu, Georgios Kementzidis, Karen Cho, Sebastian Ramirez Villarreal, Yuefan Deng. *TMLR*, *accepted with minor revision* (2026). ([OpenReview](#)) ([arXiv](#))
- [j.2] **Multistep Generative Backmapping of Coarse-Grained Structures.** Georgios Kementzidis, Erin Wong, John Nicholson, Ruichen Xu, Yuefan Deng. *Computer Physics Communications*, **327**, 110286 (2026). ([DOI](#)) ([Elsevier Share Link](#))
- [j.3] **Dynamic Schwartz–Fourier Neural Operator for Enhanced Expressive Power.** Wenhan Gao, Jian Luo, Ruichen Xu, Yi Liu. *TMLR* 2025. ([OpenReview](#))
- [j.4] **Coordinate Transform Fourier Neural Operators for Symmetries in Physical Modeling.** Wenhan Gao, Ruichen Xu, Hong Wang, Yi Liu. *TMLR* 2024. ([OpenReview](#))

### Peer-Reviewed Workshops

- [w.1] **APOD: Adaptive PDE-observation diffusion for physics-constrained sampling.** Ruichen Xu, Haochun Wang, Georgios Kementzidis, Chenhao Si, Yuefan Deng. *ICML 2025 Workshop on Assessing World Models (Poster)*. ([OpenReview](#))
- [w.2] **RL-QESA: Reinforcement-Learning Quasi-Equilibrium Simulated Annealing.** Ruichen Xu, Kai Li, Haochun Wang, Georgios Kementzidis, Wei Zhu, Yuefan Deng. *2nd AI for Math Workshop @ ICML 2025 (Poster)*. ([OpenReview](#))
- [w.3] **An Iterative Framework for Generative Backmapping of Coarse-Grained Proteins.** Georgios Kementzidis, Erin Wong, John Nicholson, Ruichen Xu, Yuefan Deng. *ICML 2025 GenBio Workshop (Poster)*. ([OpenReview](#))

### Preprints

- [p.1] **HP-JEPA: Hierarchical Partitioning for Multi-Resolution Graph Joint-Embedding Predictive Learning.** Ruichen Xu, Jingxiang Qu, Wenhan Gao, Jiaxing Zhang, Linsey Pang, Ravid Shwartz-Ziv, Yann LeCun, Yuefan Deng. *arXiv:2608.00491*, 2026. ([arXiv](#))
- [p.2] **Can Data-Driven Dynamics Reveal Hidden Physics? There Is a Need for Interpretable Neural Operators.** Wenhan Gao, Jian Luo, Fang Wan, Ruichen Xu, Xiang Liu, Haipeng Xing, Yi Liu. *arXiv:2510.02683*, 2025. ([arXiv](#))
- [p.3] **The Impact of Move Schemes on Simulated Annealing Performance.** Ruichen Xu, Haochun Wang, Yuefan Deng. *arXiv:2504.17949*, 2025. ([arXiv](#))

### Manuscripts Under Review

- [s.1] **JENO: Full-Field Latent Prediction for Sparse Inverse PDE Inference.** Ruichen Xu et al. *ICLR* 2027.
- [s.2] **SCOPE: Observation-Conditioned Full-Target Prediction for Sparse PDE Inference.** Ruichen Xu et al. *ICLR* 2027.

## TEACHING EXPERIENCE

- **Stony Brook University – Department of Applied Mathematics & Statistics (AMS)** *Aug 2022 – Present*  
*Teaching Assistant & Instructor* Stony Brook, NY, USA
  - **Teaching Assistant (Aug 2022 – May 2025):** AMS 310 (F'22, S'23), AMS 326 (S'24), AMS 510 (F'24; led recitations), AMS 595 (F'24), AMS 528 (S'25), AMS 502 (S'25); ran discussions, grading, and office hours.
  - **Instructor (Jan 2023 – Present):** AMS 394 (W'23, W'24, Su'24, Su'25), AMS 326 (Su'25), AMS 361 (W'26); designed/delivered lectures, created and graded assessments, and mentored students.

## TALKS & PRESENTATIONS

- **Lightning Talk – Diffusion4PDE**, IACS Annual Meeting, 2025; **Lightning Talk – Active Learning for Neural Operators**, IACS Annual Meeting, 2024; **Student Seminar Speaker – AI for PDEs**, IACS Student Seminar, 2025.

## ACADEMIC SERVICE

- **Gold Reviewer**, *ICML* 2026. Conference service: *ICLR* 2026/2027; *NeurIPS* 2026; *IJCNN* 2025/2026; *KDD AI4Sciences* Track 2026/2027 and Research Track 2027; *AAAI* 2027 Program Committee.
- Journal reviewer: *TMLR* 2026; *TNNLS* 2025; *Neurocomputing* 2026. Workshop reviewer: *AI for Math at ICML* 2025.

## REFERENCES

### Prof. Yuefan Deng

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Stony Brook University  
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Relationship: *Ph.D. Advisor*